

VARUN BHADURGATTE NAGARAJ

AUSTIN, TX | (512) 212-6269 | VARUNNAGARAJ2009@GMAIL.COM

EDUCATION

11th Grade Student

Round Rock High School — 23-27

STEM & Business Industry

Coursework: AP Calculus BC, AP Computer Science A, AP Research, Digital Electronics, AP Physics I, Advanced Computer Science I

SKILLS

Programming

Python, HTML, CSS, Java, C/C++,
JavaScript

Frameworks & Tools

Flask, React, Supabase, Selenium,
Tailwind CSS, Git, Requests,
JSON, BeautifulSoup

Robotics Platforms

VEX V5 (PROS, LemLib), FTC SDK
(Java)

CERTIFICATIONS

Autodesk Certified User: Inventor

Certiport

2024

Microsoft Office Specialist: Word, Excel, PowerPoint

Certiport

2023

NOCTI Business Management & Administration

Certiport

2025

PROFESSIONAL OVERVIEW

Aspiring software engineer and robotics developer with experience in full-stack systems, AI applications, and simulation platforms. Led projects scaled to 50,000+ users and built production-ready tools using React, Flask, and Supabase. Strong in systems design, real-time data, and performance optimization, with a focus on scalable, user-centric solutions.

WORK EXPERIENCE

Project Team Lead

Virtual Robotics Simulator | June 2025 – Present

- Led development of a full-stack robotics simulation platform, replacing legacy VEX tooling with integrated CAD, transpiler, compiler, and real-time simulation.
- Built detailed CAD models of competition fields and environments, enabling accurate physics-based simulation and testing.
- Developed system for importing custom VEX robot designs into the simulator, supporting modular configurations and realistic interaction.
- Architected a custom transpiler converting high-level code (FTC Java, VEXcode) into VEX-compatible commands for cross-platform execution.
- Scaled platform to 50,000+ monthly users through documentation, feature iteration, and community outreach.
- Managed cross-functional teams across software, firmware, and UX, driving system architecture and product direction.

Summer Intern

Virtual Robotics Simulator | May – April 2025

- Debugged and optimized robotics simulation codebases and assisted with CAD design and mechanical modeling.
- Led internal testing and documentation while mentoring FTC teams in autonomous programming and control systems.
- Developed React Native web and mobile interfaces for simulator access and telemetry review.

WORK EXPERIENCE CONT.

Founder & Lead Developer

HallHop | November 2024 – Present

- Founded and led development of a full-stack Chrome Extension and Flask-based platform replacing paper hall passes with real-time student tracking across campuses.
- Architected secure backend with Supabase, JWT authentication, AES-encrypted credential storage, Redis session management, and FERPA-conscious row-level security.
- Engineered high-performance data ingestion pipeline by replacing Selenium with Requests/BeautifulSoup, improving API response speed by 7×.
- Built automated outreach system that researched school districts, generated targeted contact lists, and deployed personalized email campaigns to drive adoption.
- Scaled product to 4,000+ student users, designing for multi-school deployment and 10,000+ user scalability.
- Led a team of 4 developers, coordinating backend, frontend, and product decisions while collaborating with school administrators and IT staff for deployment.
- Developed real-time logging and analytics system for student check-in/out events, enabling operational visibility for school staff.

Lightspeed/IT Director

Round Rock High School CO-OP | March 2025 – Present

- Directed a 35-member student IT organization across development, QA, and outreach, driving technical strategy and operational improvements for the school's e-commerce platform.
- Engineered and deployed a custom iOS inventory management app with barcode scanning and real-time database sync, significantly improving stock accuracy and reducing manual errors.
- Managed and extended an Ecwid-based storefront integration (Vercel + serverless APIs), implementing custom checkout logic, pricing rules, and scheduling constraints.
- Built client-side and serverless features including dynamic checkout validation, automated surcharge calculation, and runtime UI customization for improved user experience.
- Operated and scaled an e-commerce platform with 15,000+ monthly users and consistent revenue generation, optimizing workflows and system reliability.
- Led cross-functional coordination with school administrators and principal to deploy technical solutions aligned with operational and accessibility needs.
- Organized technical outreach events and promotions, leveraging automation and web tooling to drive engagement and adoption.

Lightspeed/IT Director

Round Rock High School CO-OP | January 2025 – March 2025

- Developed automation workflows to rebuild 400+ product listings with updated photography, formatting, and descriptions, improving catalog accuracy.
- Integrated Stripe checkout with Apple Pay and Google Pay, improving payment accessibility.
- Enhanced frontend UX with CSS animations to improve interaction and flow.
- Collaborated with administrators, CTE coordinators, and special education staff to support a delivery-based ordering system, reducing disruptions and increasing inclusivity.
- Assisted with backend inventory verification, product organization, and workflow improvements to enhance efficiency.

WORK EXPERIENCE CONT.

Software Engineer & Shareholder

Transforze | August 2025 – Present

- Designing an AI-native, multi-tenant CRM using React, TypeScript, Expo, Supabase, and SQL with real-time SSE streaming.
- Building agent-driven workflows with deterministic, auditable behavior using RLS, Supabase Edge Functions, and MCP-based AI agents.
- Integrated local LLMs via Ollama and implemented vector database-backed memory for persistent agent context.
- Developed MCP server integrations for Google services (Gmail, Calendar, Drive, Docs) enabling autonomous agent actions.
- Contributing to native CRM features including company creation, user management, and scalable multi-tenant architecture.

Machine Learning Intern

LPL Financials | March 2025 – Present

- Developed a conversational AI chatbot for financial client interactions, improving accessibility and response efficiency.
- Assisted in model design, training, testing, and performance evaluation for NLP-based systems.
- Collaborated with project managers and engineering teams, gaining experience in software development workflows and quality control processes.
- Participated in team meetings and product discussions to understand real-world deployment of AI systems in a financial environment.

CLUBS & ORGANIZATIONS

Software Engineer & Shareholder

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